

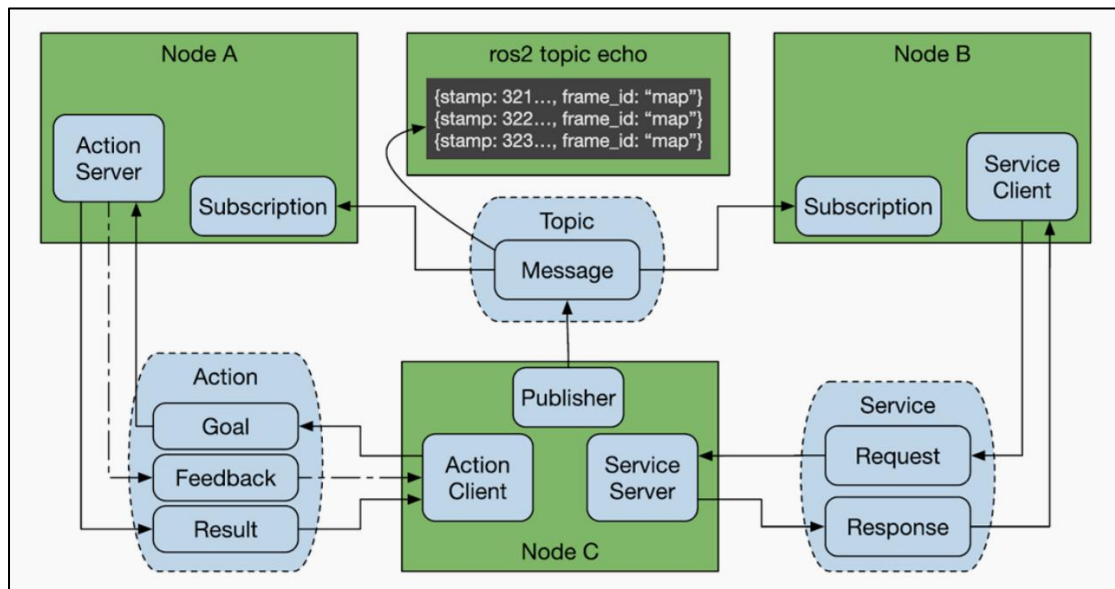
# Lesson 11 ROS2 Communication Interface

## 1. Communication Interface Introduction

In the ROS system, regardless of topics, services or actions, they all rely on an significant concept --communication interface.

Communication isn't a monologue but rather an exchange between two or more parties. What is being exchanged? To ensure mutual understanding, we can define a standardized structure for the data being transmitted, and this is what we call a communication interface.

Interfaces help reduce dependencies between programs, making it easier for us to use others' code and for others to use our code. This is the core goal of ROS, which is to reduce redundant efforts.

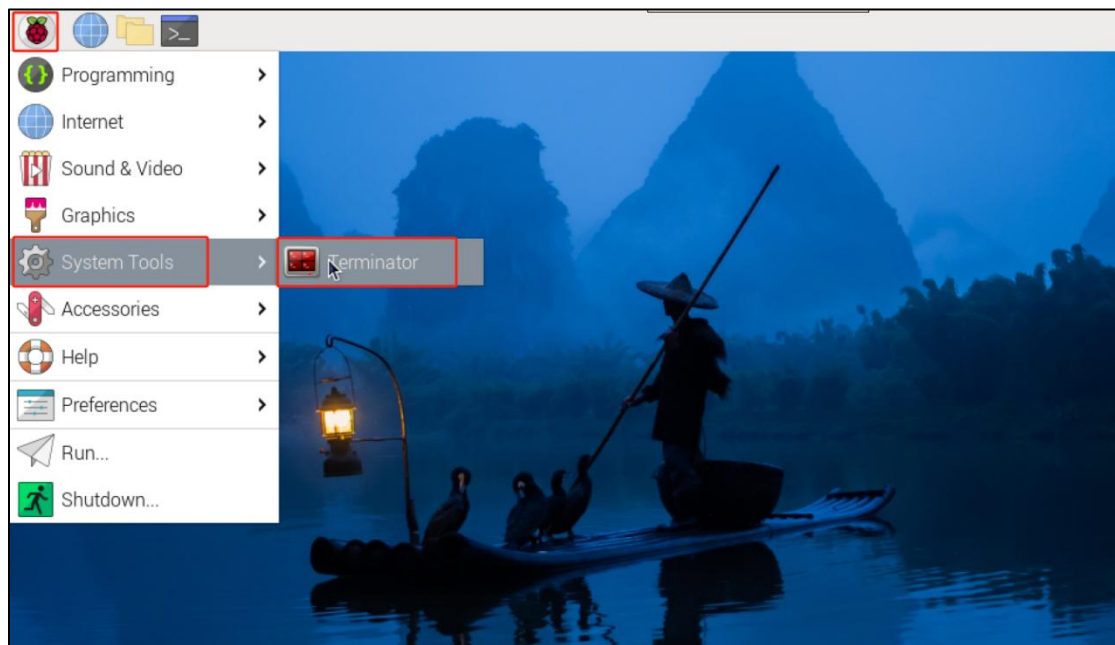


ROS has three commonly used communication mechanisms, namely topics, services, and actions. Through the interfaces defined by each communication mechanism, various nodes can be interconnected.

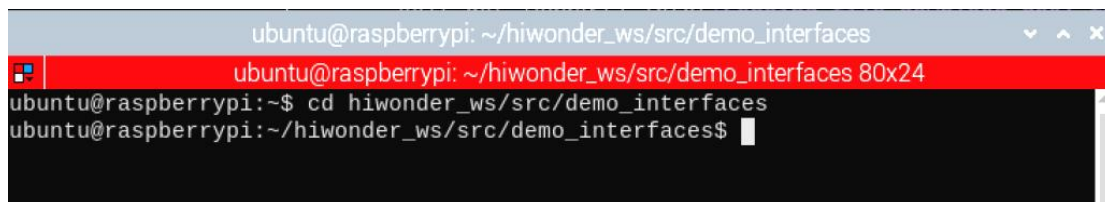
## 2. Custom Interface

In the Lesson 8 ROS Topic, Lesson 9 Service Instruction, and Lesson 10 ROS Action, the custom interface package named `demo_interfaces`. Within this package, three custom interfaces are created, which are `String.msg`, `FileDownload.action`, and `AddInts.srv`.

- 1) Click  and select "System Tools -> Terminator" in sequence.



- 2) Enter the command "`cd hiwonder_ws/src/demo_interfaces`" to switch to the ""demo\_interfaces" custom interface package.



- 3) Enter "`tree -L 1`" and press Enter to view the root directory of the package.

```
ubuntu@raspberrypi:~/hiwonder_ws/src/demo_interfaces$ tree -L 1
.
├── CMakeLists.txt
├── action
├── include
├── msg
├── package.xml
├── src
└── srv

5 directories, 2 files
```

The demo\_interfaces package has action, msg and srv folders for storing custom interface files. Following provides an explanation for each folder:

| Folder Name | Instruction                                         |
|-------------|-----------------------------------------------------|
| action      | The .action file for storing action communication.  |
| msg         | The .msg file file for storing topic communication. |
| srv         | The srv file file for storing service communication |